

Note:

- Instead of using array maybe use classes and attributes
- > should be more performance optimized

--> class: dataset

---> attributes: e.g. dataset.accelerationX1

- Logging optional depending on uC performance

- All Status changes should be communicated with telemetry

- Instead of Booleans use Status byte and check for status byte

Statusbyte structure (should be send to telemetry at the beginning of each loop):

- 1 byte (char)
- States:
 - 0000 0000 (0): Pad Idle
 - 0000 0001 (1): Armed
 - 0000 0010 (2): Liftoff detected
 - 0000 0011 (3): Booster Burnout/Coasting
 - 0000 0100 (4): Apogee detected
 - 0000 0101 (5): Pyro1 signal sent (drogue)
 - 0000 0110 (6): Pyro2 signal sent (drogue)
 - 0000 0111 (7): Pyro3 signal sent (drogue)
 - 0000 1000 (8): Pyro4 signal sent (drogue)
 - 0000 1001 (9): Drogue deployment detected
 - 0000 1010 (10): Pyro1 signal sent (main)
 - 0000 1011 (11): Pyro2 signal sent (main)
 - 0000 1100 (12): Pyro3 signal sent (main)
 - 0000 1101 (13): Pyro4 signal sent (main)
 - 0000 1110 (14): Main deployment detected
 - 0000 1111 (15): Landed

- IMU:
 - 3 axis acceleration
 - 3 axis angular velocity
- 2x pressure sensors:
 - barometric pressure values
- 2x accelerometers:
 - 3 axis acceleration
- GNSS altitude (received from sensorics)
- Continuity checks (send to telemetry)

=> Array structure: see size of each datapoint and use predetermined array structure e.g. array[0] to array[3] first IMU integer...

acceleration pointing in right direction and cleared of gravity?

Set boolean to true
-> needed later on

rotation threshold up to debate
-> indicated rocket is pitching which (ideally) only happens at apogee

Done by reading photoresistors

This is where you start celebrating, this is happy times

This is where you get off your lazy ass and start looking for that thang

